Connor Nieh

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Design Portfolio: https://connornieh.github.io/med-device-portfolio/

EDUCATION

Duke University, M.Eng in Medical Technology Design *Expected Dec. 2026*
- Coursework: Advanced Design and Manufacturing, Design Health, Signal Processing, Machine Learning and Imaging, Quality Management Systems

University of California, Riverside, B.S. Bioengineering Sept. 2021 – June 2025
- Coursework: Bioinstrumentation, Biomaterials, Biomechanics, Circulation Physiology, C++, Electrical Engineering, Optical Imaging, Statistics, Tissue Engineering, Quantitative Physiology
- Awards: Cum Laude, Dean's Honors List, Chancellor's Honors List, 3.76 GPA

MEDICAL DEVICE DEVELOPMENT EXPERIENCE

Graduate Research Assistant, BIOS Lab – Duke University Sept. 2025 – Present
- Transforming a quantitative phase imaging system into a low-cost diagnostic device for sickle cell disease, enabling point-of-care clinical application in low access healthcare environments
- Implementing machine learning workflows to process and analyze high-throughput flow cytometry experiments
- Working in a clean room to produce novel microfluidic chips and develop a repeatable manufacturing protocol

Design Health Project, Duke Design in Healthcare Series Aug. 2025 – Present
- Engaging in ethnographic research and clinical shadowing to define unmet needs in cardiothoracic operating rooms
- Driving early-stage prototyping, design documentation, and bench-testing of a clinically-driven surgical device
- Utilizing root cause analysis and MECE organization to translate hundreds of needs into an engineering solution

Microscope Integrated Instrumentation for OCT Project, ONI Lab - UCR Sept. 2024 – June 2025
- Prototyped and validated a device to integrate an optical coherence tomography (OCT) system into an upright microscope to enable simultaneous, multi-modal imaging of tissue structures
- Optimized imaging clarity and OCT system performance by designing precise optomechanical assemblies to collimate and align optical components in a custom housing unit

Medical Device Development Intern, Keck Graduate Institute June 2023 – July 2023
- Co-designed and prototyped a medical device to promote neuroplasticity in dementia patients, integrating CAD, electrical engineering, and Python programming skills to deliver a functional, patient-focused product
- Leading an interdisciplinary team through regulatory, quality management system, and R&D pathways

RESEARCH EXPERIENCE

Undergraduate Researcher, TIME Lab - UCR Oct. 2023 – June 2025
- Engaged in disseminative research by investigating stress tolerance of genetically engineered stem cell therapies and presenting findings on ASC toxic responses at the UCR Undergraduate Research Symposium
- Created an adaptable, multi-cell proliferation model of cdc2 and cyclin interactions in MATLAB

Summer Biomedical Intern, Terasaki Institute for Biomedical Innovation June 2024 – Sept. 2024
- Led parallel projects in iPSC characterization, 3-D cell modeling, and drug assay development, accelerating experiment throughput and supporting precision medicine studies in oncology
- Developed and documented over 5 tissue processing, organoid culturing, cell harvesting, and drug dosing protocols
- Presented research on novel disease modeling and cell-based diagnostic applications of organoid technologies

SKILLS

R&D: CAD Modeling, Design Documentation, DFM, DHF Documentation, Ethnographic Research, FEA, Iterative Prototyping, MECE Analysis, Root Cause Analysis, Validation Testing, QMS, 3-D Printing

Software: Arduino Programming, CB-Docking, COMSOL Multiphysics, Fusion 360, Google Suite, Image Reconstruction, Kivy, MATLAB, Microsoft Suite, On-Shape CAD, Python, SOLIDWORKS, 3-D Slicer

Laboratory: Confocal Microscopy, Disease Modeling, DNA/RNA Isolation, ELISA, Fluorescence Imaging, Microfluidic Fabrication, OCT Imaging, Optics, Organoid Characterization, PCR, Quantitative Phase Imaging, 2-D/3-D Cell Culture

EXTRACURRICULARS

International Society of Pharmaceutical Engineering Mar. 2023 – June 2025
Vice President, University of California, Riverside
- Coordinated member participation in professional workshops and networking events with industry leaders